THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–11) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A person's starch digestion starts in the mouth. If saliva is absent, the digestion of: 1

(a) starch is delayed **(b)** water stops
(c) fats stop **(d)** protein stops

2. In the variegated leaf, the white parts are tested with iodine. They: 1

(a) turn blue-black **(b)** remain brown **(c)** turn green **(d)** turn red

3. A ray strikes the mirror surface making an angle of 25° with the surface. The angle of reflection is: 1

(a) 65° **(b)** 25° **(c)** 50° **(d)** 90°

4. If the Earth's axis were not tilted, we would have: 1

(a) no seasons **(b)** no night **(c)** no day **(d)** more seasons

5. In $H_2SO_4 + 2\,NaOH \longrightarrow Na_2SO_4 + 2\,H_2O$, the salt formed is: 1

(a) sodium carbonate **(b)** sodium chloride **(c)** sodium nitrate **(d)** sodium sulphate

6. An electrician wears rubber gloves because they: 1

(a) are insulators **(b)** are costly **(c)** are heavy **(d)** are conductors

7. When sulphur is burnt and gas passed into water, the solution turns: 1

(a) no change **(b)** red litmus blue **(c)** blue litmus red **(d)** green

8. In the experiment, the nail in boiled water with oil does not rust because: 1

(a) no water **(b)** no iron
(c) no air reaches it **(d)** too much oil

9. Why do adolescents have a good appetite? 1

(a) fast growth needs more energy **(b)** they are old
(c) they are sick **(d)** they are bored

10. During land breeze, warm air rises over the: 1

(a) desert **(b)** land **(c)** sea **(d)** mountain

11. A bus travels 60 km at 30 km/h. The time taken is: 1

(a) 2 h **(b)** 30 h **(c)** 90 h **(d)** 0.5 h

12. During vigorous exercise, the breathing rate: 1

(a) decreases **(b)** stops **(c)** increases **(d)** stays same

13. Which plant cell part controls the opening of stomata? 1

(a) phloem **(b)** guard cells **(c)** xylem **(d)** root hair

14. The shadow of a pole is shortest at: 1

(a) noon **(b)** evening **(c)** morning **(d)** night

15. A full moon rises at about: 1

(a) sunrise **(b)** sunset **(c)** noon **(d)** midnight

16. Why does baking soda taste bitter? 1

(a) it is acid **(b)** it is a base **(c)** it is a salt **(d)** it is sugar

17. A student wants to test an object for conduction. Which item completes the circuit tester? 1

(a) only wires **(b)** a cell, bulb and wires

(c) only a switch **(d)** only a bulb

18. Why is silver used in mirrors and in jewellery? 1

(a) it is gaseous **(b)** it is brittle **(c)** it is lustrous **(d)** it is dull

19. In making crystals, why is the solution filtered while hot? 1

(a) to remove insoluble impurities **(b)** to remove all water

(c) to make it solid **(d)** to make it blue

20. The pancreas is in the: 1

(a) abdomen **(b)** chest **(c)** brain **(d)** neck

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Why are cells not connected to opposite polarity in series? 2

22. Give two examples where metal oxides are used as bases. 2

23. Why are crystals formed when a hot saturated solution is cooled? 2

24. What are the benefits of regular exercise for teenagers? 2

25. Why are cooking pots with copper bottoms common? 2

26. Explain why speed of a cyclist uphill is less than downhill. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Give reasons: 3

(a) pans are made of metal and have plastic handles

(b) the tank for hot water is covered with a wool jacket.

28. A man walks to a market 3 km away at 5 km/h and returns at 3 km/h. Find the total time. 3

29. What are the effects of smoking on the respiratory system? 3

30. Explain how stomata open and close. 3

31. Why do we use a mirror at the corner of a blind road? 3

OR

Describe the phenomenon of midnight Sun. 3

32. Why should chemicals in the lab be labelled? 3

OR

Describe a safe way to disconnect a plugged-in device. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Explain how the respiratory organs differ in humans, fish, insects, earthworms and amoeba. 5

34. Describe the experiment on transpiration with a polythene bag and the precautions. 5

35. Explain how a solar and a lunar eclipse occur using the ideas of shadow. 5

36. Explain why we always see the same face of the Moon. 5

OR

Describe the full activity of neutralisation of hydrochloric acid with sodium hydroxide using litmus, with equation. 5

37. Explain the heating effect of current with a simple experiment using a nichrome wire and cells. 5

OR

Explain the formation of oxides of magnesium and sulphur, their tests and equations. 5

38. Explain the changes that occur in a car's exhaust, the engine, the paint and the iron body, and classify them. 5

OR

Explain the importance of nutrition, sleep, hygiene and physical activity for adolescents with reasons. 5

END OF QUESTION PAPER Total: 80 Marks

MASTERMIND